Michael Melichar

michaelmelichar.com | michaelrmelichar@gmail.com | linkedin.com/in/michaelrmelichar | U.S. Citizen | Clearance Eligible

Education

University at Buffalo — Buffalo, NY
Bachelor of Science in Computer Science — *Aug. 2024 – Expected Dec. 2027*

Experience

UB SEDS | Avionics Software Engineer — Aug. 2024 – Present
University at Buffalo — *Buffalo, NY*

- Wrote embedded firmware in C/C++ on an STM32 running FreeRTOS to drive solenoid valves for loading nitrous oxidizer into a hybrid rocket during ground fill operations.
- Programmed a stepper-motor driver to retract the rocket's umbilical line at launch, and built the igniter firing driver.
- Built abort and safety task handling as FreeRTOS tasks that monitor fault conditions and trigger a controlled shutdown of the fill and ignition sequence.
- Prototyped a Kalman filter in C++ for IMU-based state estimation, validated against simulated accelerometer and gyroscope data.

UB Nanosatellite Laboratory | Flight Software Engineer — Sept. 2025 – Present
University at Buffalo — *Buffalo, NY*

- Write flight software in C on NASA core Flight System (cFS), handling telemetry and command messages routed over the cFS Software Bus.
- Built test applications in C that send health packets over UART through the EyeStar radio to verify connectivity with the Globalstar satellite network.
- Integrated message passing between the flight computer and an onboard NVIDIA Jetson GPU using the lab's payload communication API.
- Bench-tested sensor and radio hardware on Linux-based flight computers, debugging serial links and confirming packet integrity end to end.
- Tracked down repeated flash failures on a TI TMS570LC43 Hercules board to the reset button clearing only the error LED and not the caches, then fixed it by clearing the registers by hand.

Linde | Software Engineering Intern — May 2026 – Present
Buffalo, NY

- Build a customer-facing web app in C# and Blazor that visualizes live and historical readings from cryogenic freezers, so customers can track temperature and fill levels directly.
- Write data pipelines that pull historical weather data from public web APIs into Microsoft Fabric, feeding a forecasting model covering 8,000+ plant locations.
- Model and structure the ingested data in Microsoft Fabric to support reporting and downstream analysis.

Projects

Create: Civilization (Founder & Owner) | *Java, Kotlin, Python, React, SQLite* — June 2024 – Present

- Founded and run a modded Minecraft server network that has grown to 680+ users at 99.8% uptime, including load balancing and performance monitoring.
- Raised concurrent player capacity from 20 to 60 by making authentication, the join pipeline and chunk loading async with Java mixins, and by pre-generating the world map.
- Built and maintain the community website in React and Discord moderation and alert bots in Python over REST APIs.
- Run the team on Scrum and wrote Python scripts for player onboarding, automated backups, and log parsing.

Technical Skills

Embedded / Systems: NASA cFS, FreeRTOS, STM32, TI TMS570, Linux, UART/Serial, GPIO, Raspberry Pi, NVIDIA Jetson
Languages: C/C++, C#, Python, Java, Kotlin, JavaScript, SQL (Postgres, MySQL, SQLite), HTML/CSS
Web / Frameworks: Blazor, .NET, React, Node.js, Flask, FastAPI
Tools / Cloud: Git, GitHub Actions (CI/CD), Docker, Microsoft Fabric, Azure, Visual Studio, VS Code, IntelliJ